



ML and FinTech

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National Yang Ming Chiao Tung University

Outline

1. About ME: <https://venteng.github.io>

2. Syllabus

- ▶ The course centers on **data analytics** within FinTech!
- ▶ Infrastructure
- ▶ Grading policies

3. FinTech

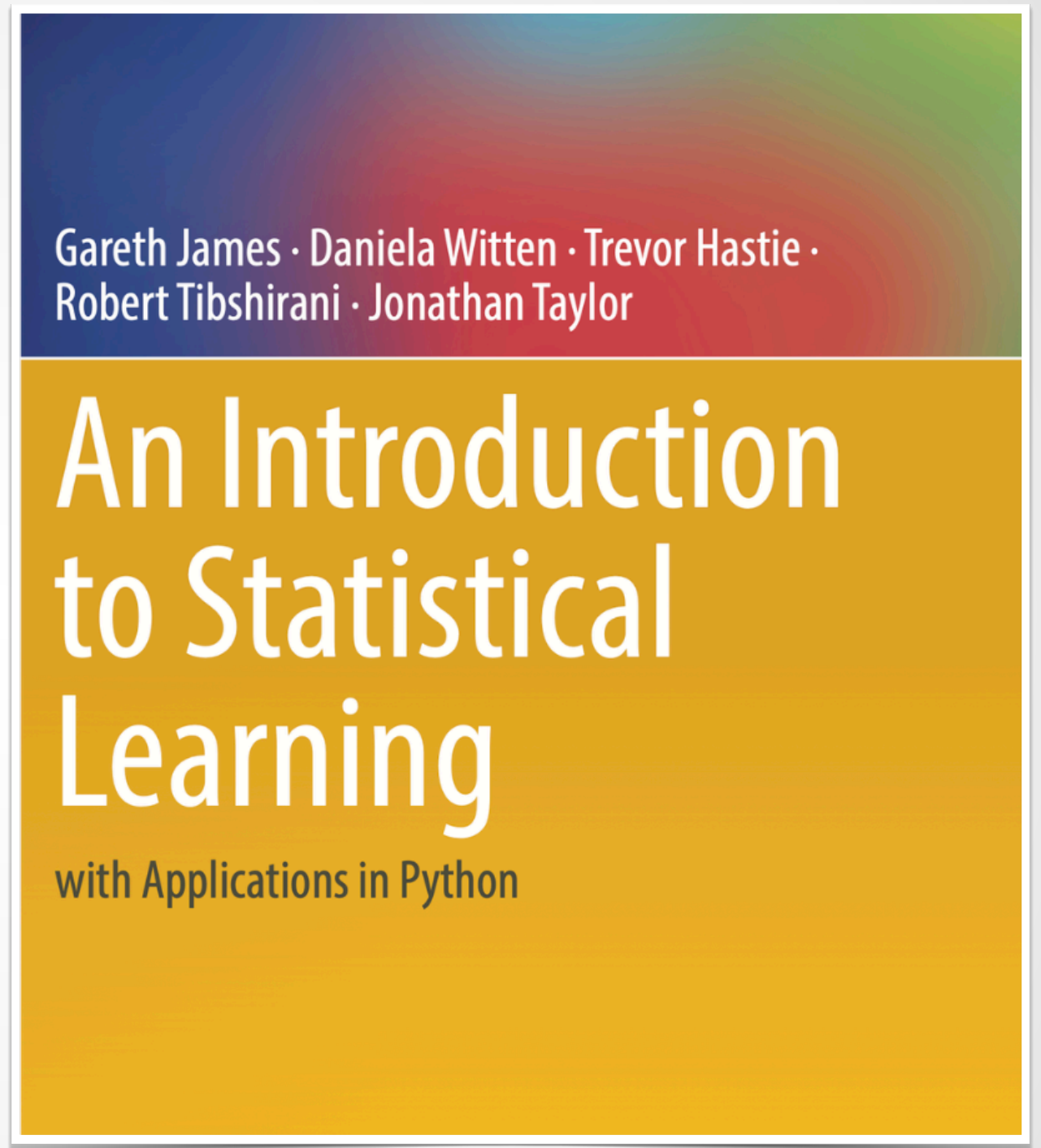
4. Stat, ML, and AI

5. More



Introduction

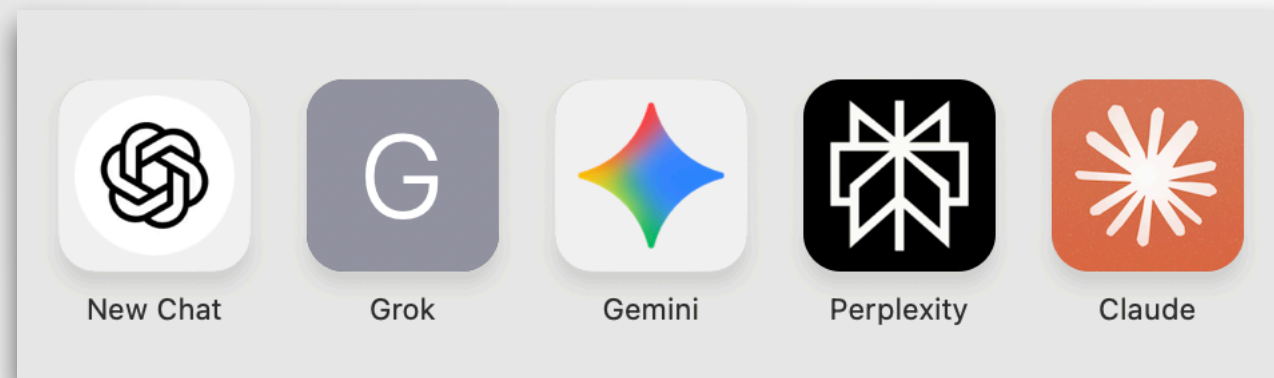
- ▣ MUST:
 - ▣ Laptop & internet: In-class exercises and presentations
- ▣ TA: 陳之怡 (Claire Chen)
Email:
claire.mg14@nycu.edu.tw



<https://www.statlearning.com/>



We embrace genAI!



- We will use cold calls during course!



Infrastructure

- ▣ E3: Scores recording, announcement
- ▣ GitHub
 - ▶ 202609-ML-FinTech
 - ▶ Your repo: 202609-ML-FinTech-[no]-[name]
 - ▶ HW/
 - ▶ Project/: README.md, logs, snapshots, data and codes
- ▣ Overleaf: Manuscripts in a professional writing style using latex
- ▣ Link to collect info for GitHub, overleaf



Grading policy

Index	Items	%	Details
1	Participation	20%	in-class exercise, presentation: HW, project, summary of the course, and others. We will use cold calls during class.
2	Project	40%	Grade based on your manuscript (written in latex). Slides needs to be generated for presentation.
3	Exam	40%	* In class and with one A4 double-sided cheating sheet!
	Total	100%	



Participation (may adjusted slightly later)

Index	%	Details
1	8%	Presentation: project (slides), MFS (slides), HW (GitHub repo0, course summary, etc.
2	12%	In-class exercise
	20%	

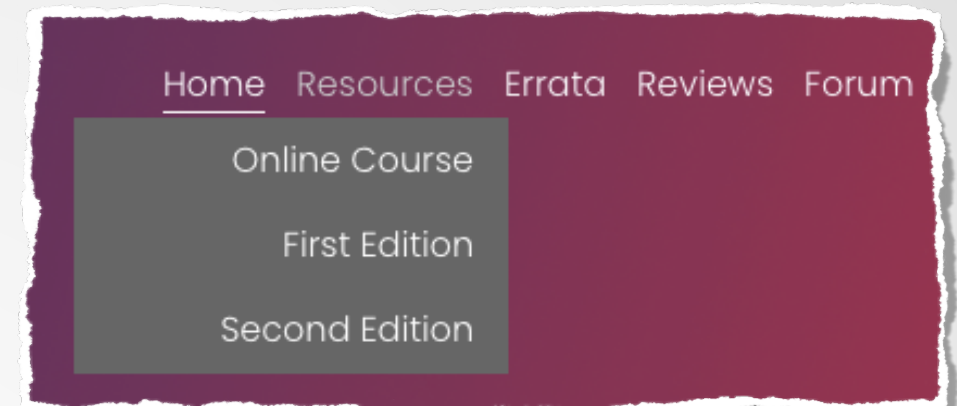


Topics

- ▣ Techniques
 - ▶ Machine Learning (supervised and/or unsupervised learning)
 - ▶ LLM
- ▣ Topics
 - ▶ Credit scoring/credit rating
 - ▶ Fraud detection
 - ▶ Trading strategies
 - ▶ Portfolio management
 - ▶ Others related to FinTech



Textbook: Introduction to Statistical Learning



The First Edition topics include:

- Sparse methods for classification and regression
- Decision trees
- Boosting
- Support vector machines
- Clustering

The Second Edition adds:

- Deep learning
- Survival analysis
- Multiple testing
- Naive Bayes and generalized linear models
- Bayesian additive regression trees
- Matrix completion

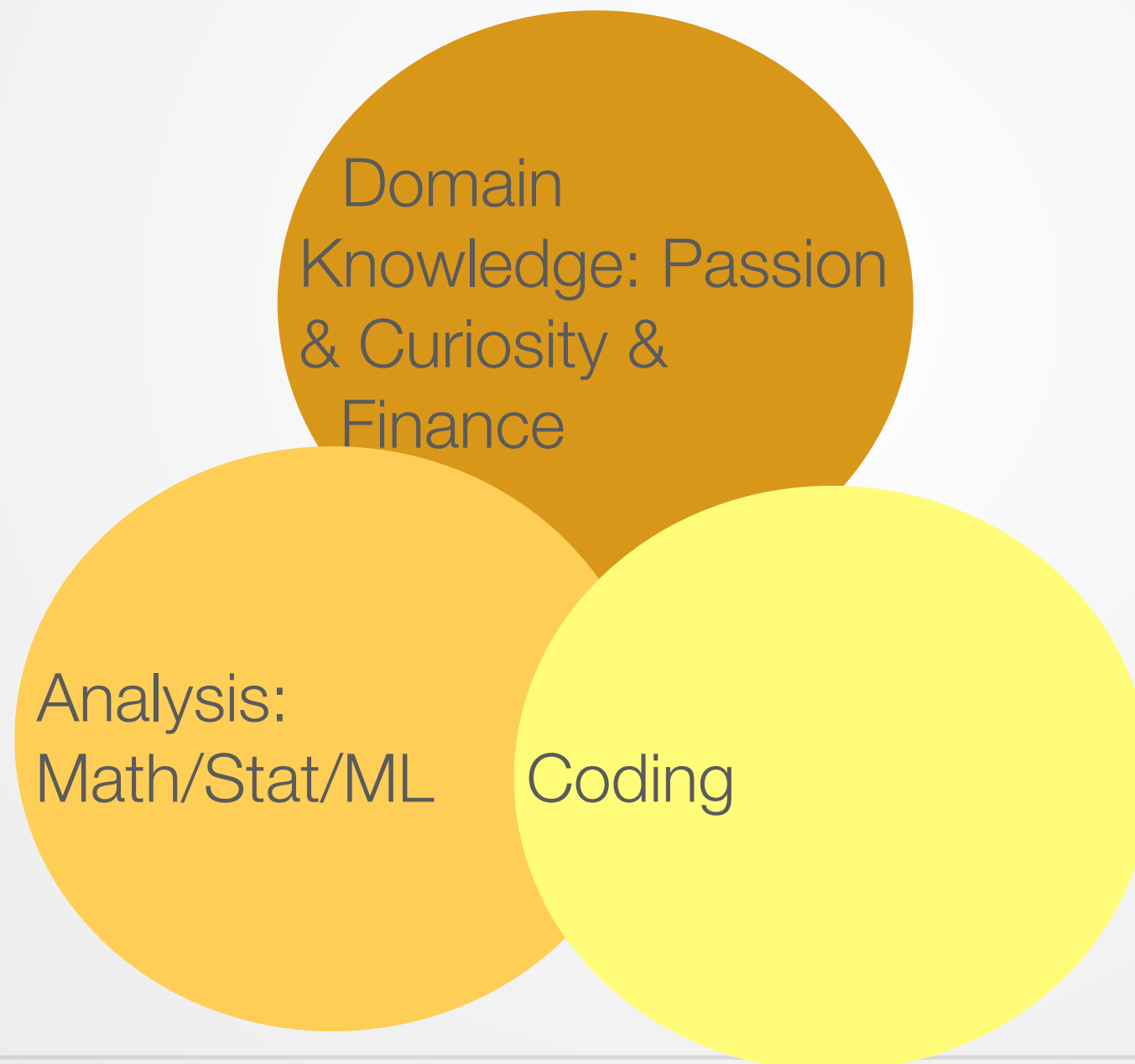
<https://www.statlearning.com/resources-second-edition>

- + **A Note About the Chapter 10 Lab**
- + **.R Files**
- + **Rmarkdown Files**
- + **Jupyter Notebook Files**
- + **Slides**
- + **Data Sets**
- + **Figures**



Presentations, presentations, presentations!

- ▣ Slides: Using my template in Canva
- ▣ Manuscript: latex editing in Overleaf



Your GitHub Repo

- ▣ README.md:
 - ▣ Link to your own GitHub page
 - ▣ Replicating a paper: Chicago style to reference a paper
 - ▣ Links to your manuscript via overleaf (shared with me at venteng@gmail.com)
 - ▣ Links to Slides via Canva (shared with me at venteng@gmail.com)
- ▣ Slides
- ▣ “Code”: Python in Jupyter Notebook
- ▣ “Data”: datasets or links to data (in README.md).



Potentials of your projects

- ▣ With modifications:
 - ▣ Thesis or dissertation
 - ▣ Journal publication



Tools

- ▣ Markdown使用說明
 - ▣ <https://hackmd.io/8nPFj8X7Rc2UhkhfjYAbkw?both>
- ▣ 哪裡可以找到資料?
 - ▣ <https://hackmd.io/LfakJmiPQCaui48zAx71xw>
- ▣ 高速運算: 國網中心
 - ▣ <https://hackmd.io/HXY75BRpRzimkWdCIAbuLw?both>
- ▣ A data competition hold at Stat, Sinica
 - ▣ <https://www3.stat.sinica.edu.tw/pds2026/>





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